Extended Case 1 — Springville Herald

Calls, subscriptions and forecasting · 307312 Applied Statistics — Masters Level

- **Start after:** Part 2 for Section A; Part 3 regression for Section B.
- **Optional advanced work:** Section C introduces trend forecasting.
- **Suggested time:** 45 minutes for A; 60 minutes for B; 30 minutes for C.
- **Business question:** How can a newspaper describe calling activity, plan telemarketing and forecast its subscription base?

Learning outcomes:

- Summarize daily variation without confusing days and months.
- Interpret a regression slope and distinguish mean-response and prediction intervals.
- Recognize temporal dependence and the limits of a straight-line forecast.

1. The circulation manager's brief

- The Springville Herald wants to grow home-delivery subscriptions and coordinate delivery capacity.
- Marketing records calling activity and telemarketing hours; circulation tracks the subscription base.
- **Source:** Berenson, Levine and Krehbiel, *Basic Business Statistics: Concepts and Applications*, ninth edition, Springville running case; Chapters 2, 13 (pp. 536–537) and 16 (pp. 689–690).
- **Data:** historical publisher case measurements; independently verified real-company origins are not established.
- **Period:** SH13 and SH16 each cover 24 months; calendar dates are not supplied in these CSVs.
- **Limit:** these files have different observational units. Do not join them by row number or assume their months align.

2. Import the three files

```
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
calls <- read.csv(file.path(data_dir, "business", "extended-herald-calls.csv"))
marketing <- read.csv(file.path(data_dir, "business", "extended-herald-marketing.csv"))
subscriptions <- read.csv(file.path(data_dir, "business", "extended-herald-subscriptions.csv"))
stopifnot(nrow(calls) == 90, nrow(marketing) == 24, nrow(subscriptions) == 24)
knitr::kable(data.frame(file = c("SH2", "SH13", "SH16"),
  observations = c(nrow(calls), nrow(marketing), nrow(subscriptions)),
  unit = c("Day", "Month: marketing", "Month: subscription base")))
```

file	observations	unit
SH2	90	Day
SH13	24	Month: marketing
SH16	24	Month: subscription base

- CSVs: `data/business/extended-herald-calls.csv`, `data/business/extended-herald-marketing.csv`, `data/business/extended-herald-subscriptions.csv`.

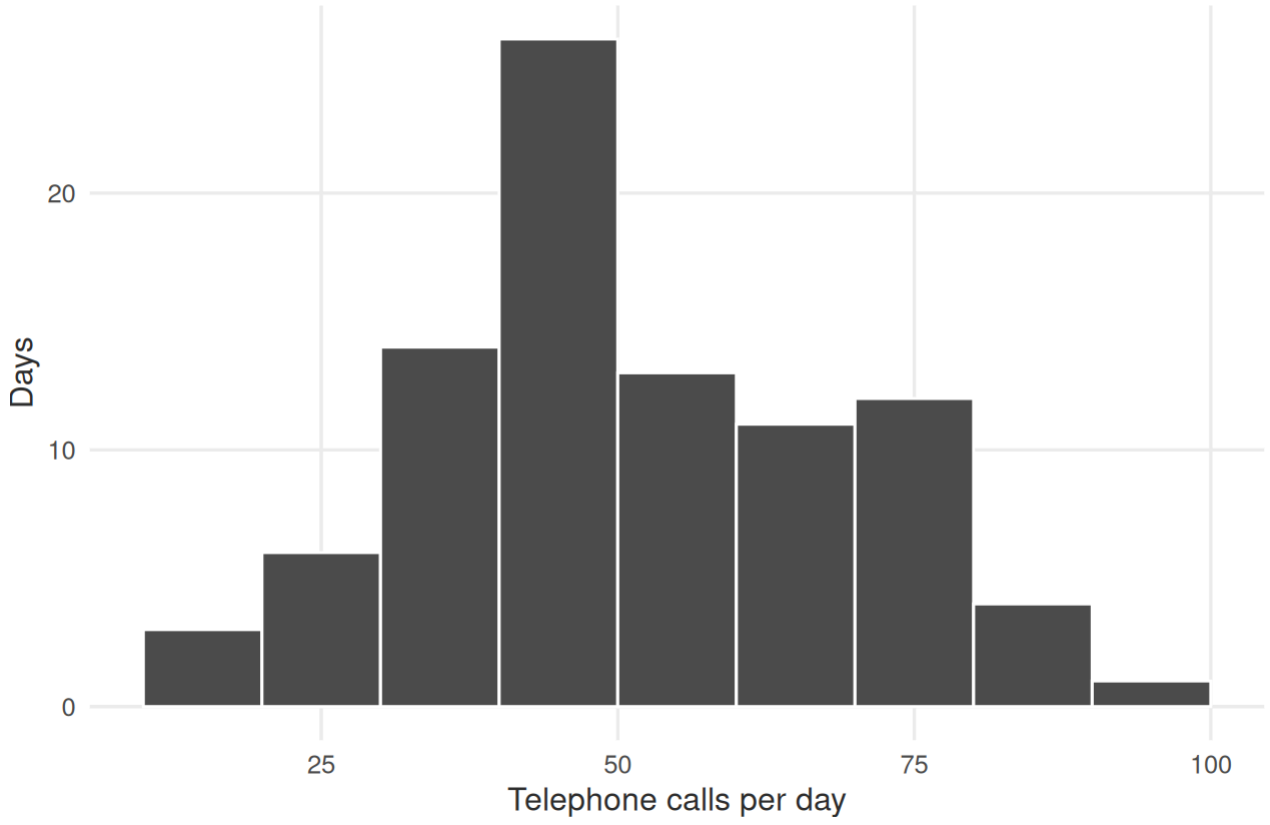
- SH2 chart-helper columns were excluded; all 90 daily call counts remain.
- `source()` loads the same R packages and plot style used in the core course.

A. Describe daily calling activity

```
call_summary <- calls %>% summarise(days = n(), mean_calls = mean(calls),
  median_calls = median(calls), sd_calls = sd(calls),
  min_calls = min(calls), max_calls = max(calls))
knitr::kable(call_summary, digits = 2)
```

days	mean_calls	median_calls	sd_calls	min_calls	max_calls
90	51.84	49	17.43	11	94

```
ggplot(calls, aes(calls)) +
  geom_histogram(binwidth = 10, boundary = 0, fill = petra_ink, colour = "white") +
  labs(x = "Telephone calls per day", y = "Days")
```



- `summarise()` reports centre and spread across days.
- The standard deviation describes day-to-day variation; it is not a standard error.
- A histogram groups call counts; its vertical axis counts days.
- **Discuss:** Would the mean alone give the circulation manager enough information?

Independent task A

1. Calculate the first and third quartiles and the IQR of daily calls.
2. Draw a boxplot and investigate any observations flagged by the 1.5-IQR rule; retain valid observations.

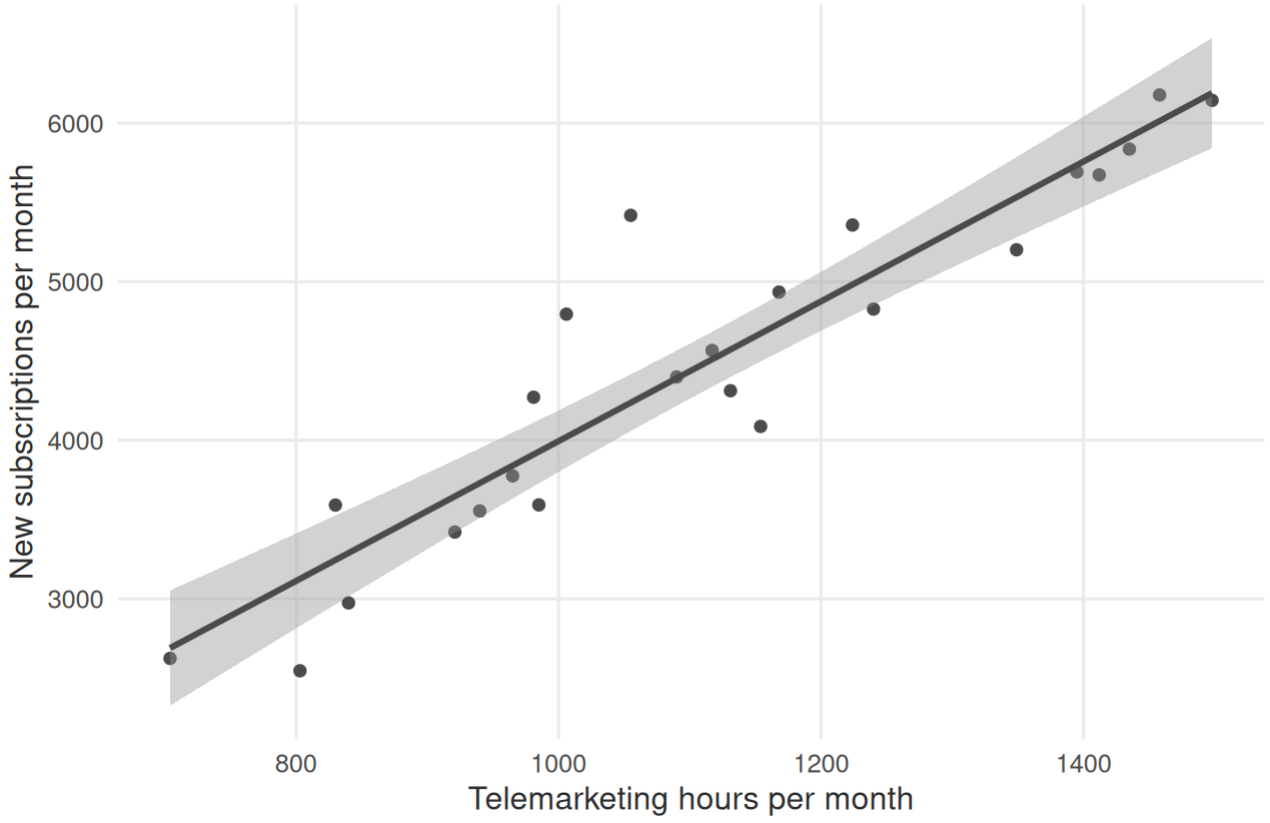
3. Explain how changing the histogram bin width affects its appearance.
4. Write three bullets about typical workload, variability and a limitation.

B. Relate telemarketing hours to new subscriptions

```
knitr::kable(data.frame(variable = c("Hours per month", "New subscriptions per month"),
  mean = sapply(marketing, mean), sd = sapply(marketing, sd),
  min = sapply(marketing, min), max = sapply(marketing, max)),
  row.names = FALSE, digits = 2)
```

variable	mean	sd	min	max
Hours per month	1112.54	226.97	704	1498
New subscriptions per month	4490.21	1077.77	2546	6177

```
ggplot(marketing, aes(hours, new_subscriptions)) +
  geom_point(colour = petra_ink) + geom_smooth(method = "lm", formula = y ~ x, colour = petra_ink) +
  labs(x = "Telemarketing hours per month", y = "New subscriptions per month")
```




```
model <- lm(new_subscriptions ~ hours, data = marketing)
summary(model)
```

```
Call:
lm(formula = new_subscriptions ~ hours, data = marketing)

Residuals:
    Min      1Q  Median      3Q     Max
-585.95 -233.67  -65.11  172.85 1181.43

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -413.8204   427.0878  -0.969    0.343
hours          4.4080     0.3765  11.709 6.36e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 409.8 on 22 degrees of freedom
Multiple R-squared:  0.8617,    Adjusted R-squared:  0.8554
F-statistic: 137.1 on 1 and 22 DF,  p-value: 6.358e-11
```

```r
confint(model)
```

```
                 2.5 %     97.5 %
(Intercept) -1299.54639 471.905508
hours           3.62724   5.188661
```

- `lm(y ~ x)` fits a straight-line relationship.
- The slope is the estimated difference in new monthly subscriptions per additional telemarketing hour.
- Observational association does not establish that increasing hours causes the predicted increase.
- Staffing, campaigns and seasonal demand may also matter.

Average response versus one future month

```r
new_month <- data.frame(hours = 1200)
knitr::kable(predict(model, newdata = new_month, interval = "confidence"), digits = 1)
```

fit	lwr	upr
4875.7	4689.3	5062.1

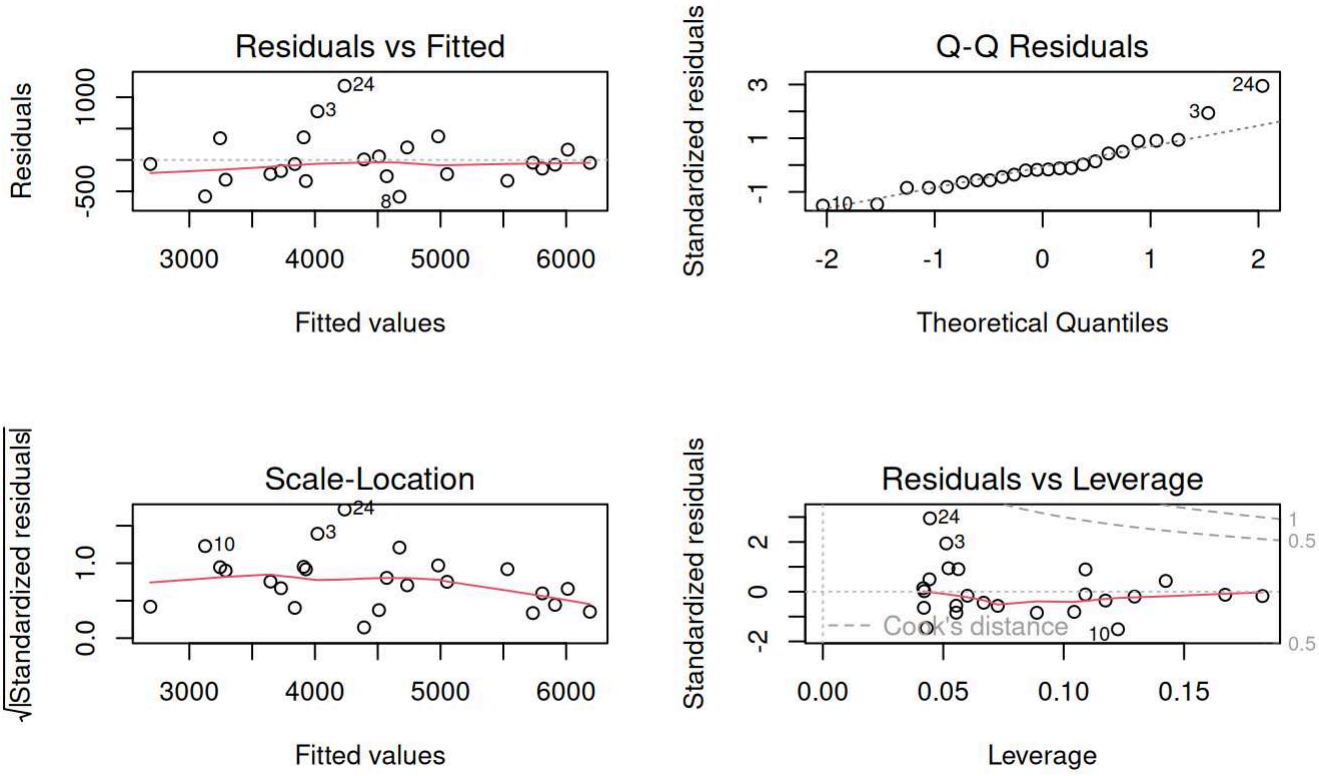
```r
knitr::kable(predict(model, newdata = new_month, interval = "prediction"), digits = 1)
```

fit	lwr	upr
4875.7	4005.7	5745.8

```r
range(marketing$hours)
```

```
[1]  704 1498
```

```r
par(mfrow = c(2, 2))
plot(model)
```




```
par(mfrow = c(1, 1))
```

- The confidence interval concerns the **mean response** at 1,200 hours under the fitted model.
- The prediction interval concerns **one future month's** response and is wider.
- Check the observed hours range before treating a prediction as interpolation.
- Residual plots help assess linearity, spread, unusual observations and normality.
- These plots alone cannot establish independence across months; the file has no month identifier.

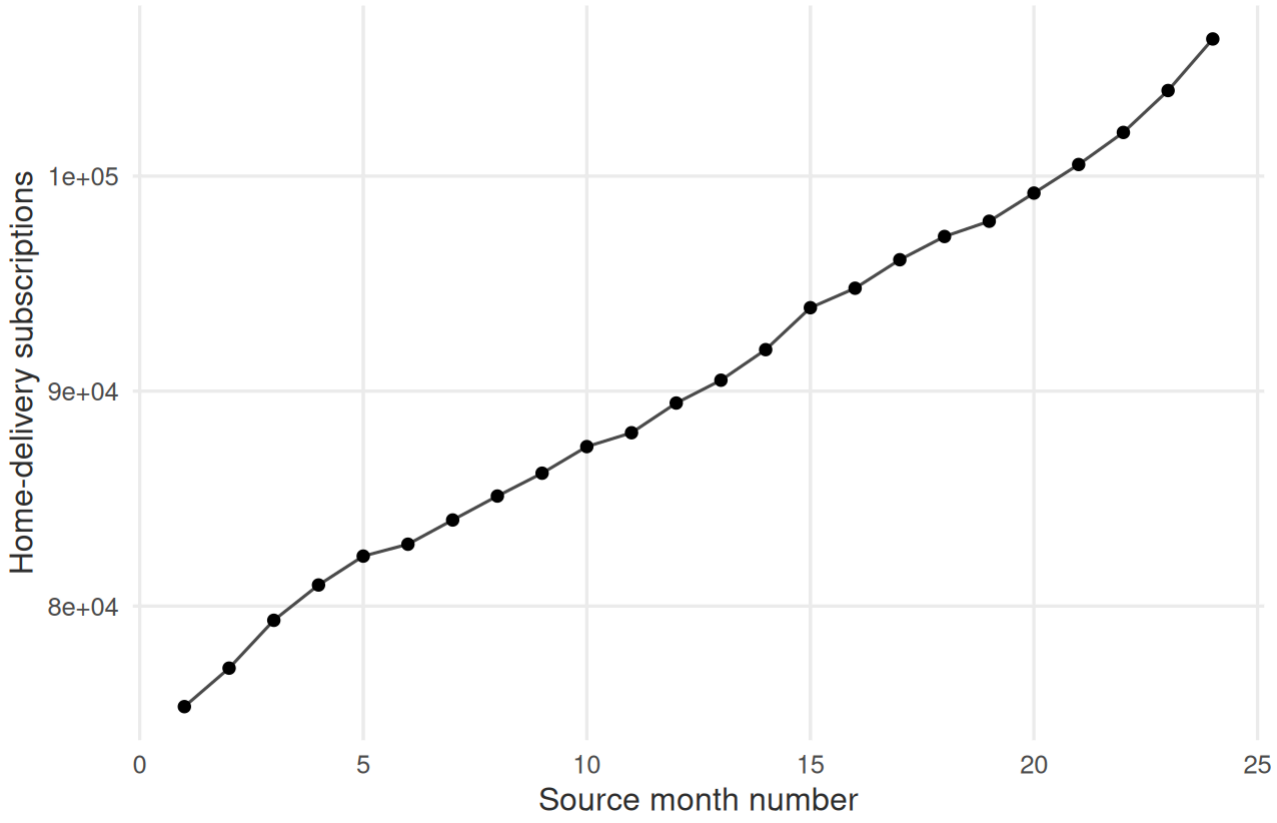
Independent task B

1. Report the fitted equation with units and interpret its slope.
2. Compare the two intervals at 1,200 hours and explain their different uses.
3. Predict at 2,000 hours; explain the additional risk using the observed hours range.
4. Identify one omitted business factor and describe how to collect it.
5. Write a 150-word staffing recommendation conditional on the model assumptions.

C. Optional advanced work: forecast the subscription base

- A **stock** is the total subscription base at a time; a **flow** is the new subscriptions during a period.
- New subscriptions do not equal net growth: cancellations and other changes also matter.
- A straight-line trend is a baseline for extrapolation, not a complete time-series analysis.

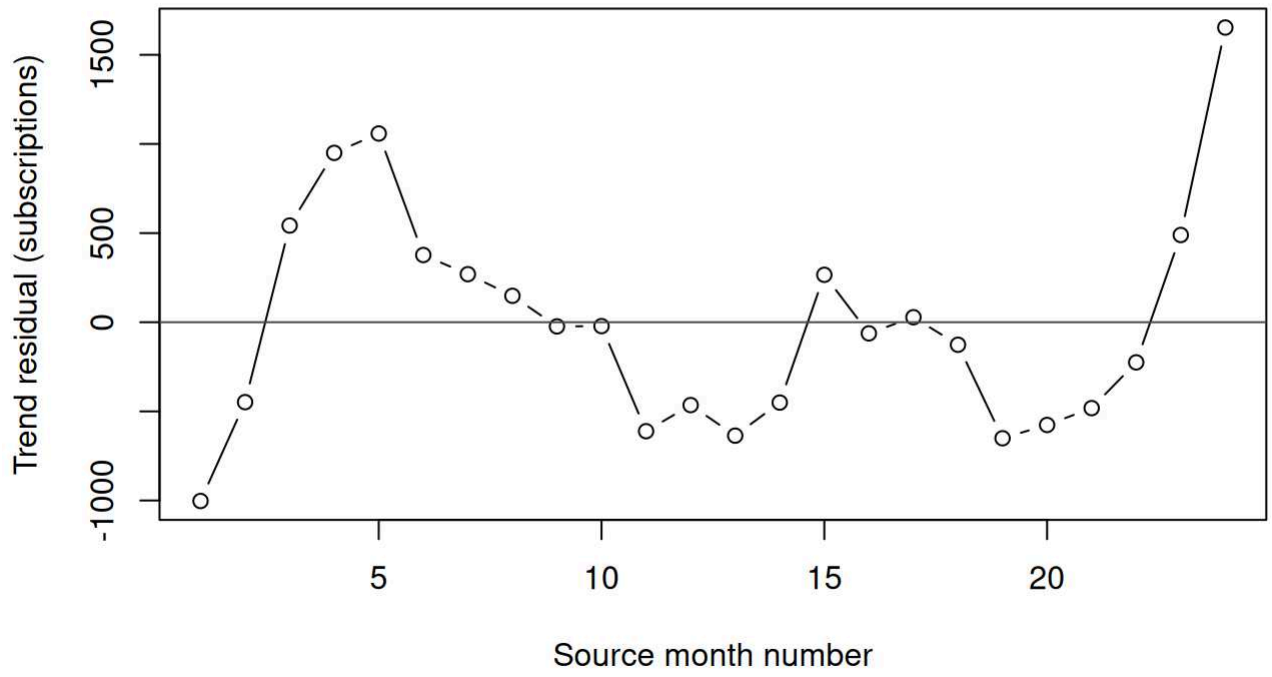
```
subscriptions <- subscriptions %>% arrange(month)
ggplot(subscriptions, aes(month, subscriptions)) +
  geom_line(colour = petra_ink) + geom_point() +
  labs(x = "Source month number", y = "Home-delivery subscriptions")
```



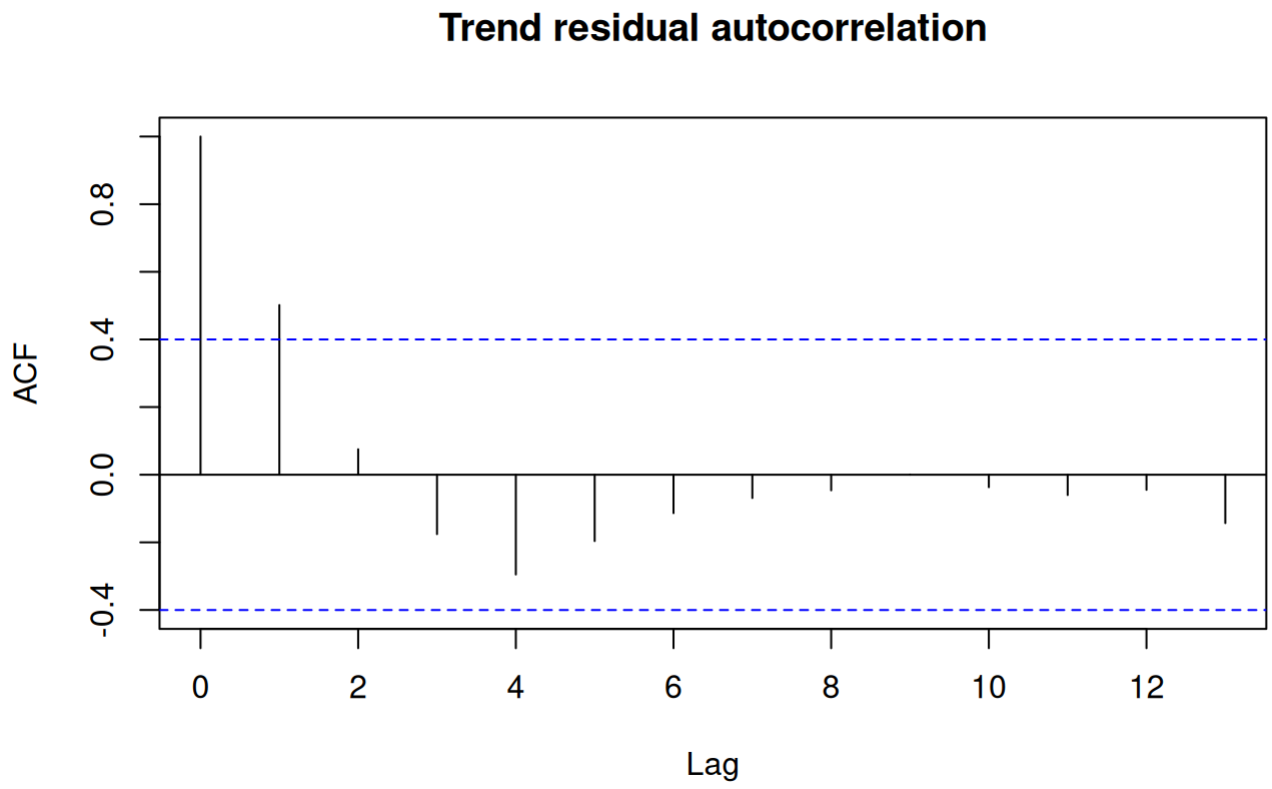

```
trend <- lm(subscriptions ~ month, data = subscriptions)
future <- data.frame(month = 25:28)
knitr::kable(cbind(future, predict(trend, future, interval = "prediction")), digits = 0)
```

month	fit	lwr	upr
25	105957	104512	107401
26	107191	105733	108649
27	108425	106953	109898
28	109660	108172	111148

```
plot(subscriptions$month, residuals(trend), type = "b",
     xlab = "Source month number", ylab = "Trend residual (subscriptions)")
abline(h = 0, col = petra_ink)
```




```
acf(residuals(trend), main = "Trend residual autocorrelation")
```



- `predict()` extrapolates the fitted line to months 25–28.
- Its nominal 95% prediction intervals assume a suitable linear model with independent, constant-variance normal errors.

- Residual patterns or autocorrelation undermine those intervals; they are not validated forecast coverage.
- With only 24 months, seasonal structure is difficult to establish reliably.

Independent task C

1. Compare the first and last subscription totals; distinguish absolute growth from percentage growth.
2. Assess the residual plots and explain whether the linear trend is an adequate baseline.
3. Write a cautious four-month capacity briefing and explain why a 12-month extrapolation would need more evidence.

Submission

- Submit the sections assigned by your instructor; the entire case is not required for every assignment.
- For each assigned section: one compact table, one labelled chart and a recommendation with a limitation.
- Keep the three datasets and their units separate.